

Epigenetic adaptation through hormone modulation in autonomous robots

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Abstract— Epigenetic adaptation provides biological organisms with the ability to adjust their physiology and/or morphology in order to meet some of the challenges posed by their environment. Recent research has suggested that this process may be controlled by hormones. In this paper, we present a model that allows an autonomous robot to develop its systems in accordance with the environment it is currently situated in. Experiments have been undertaken in multiple environments with different challenges and niches to negotiate. We have so far seen encouraging results and the emergence of unique behaviours tailored to exploiting its current environment.

I. INTRODUCTION

In biological organisms, a form of epigenetic adaptation occurs through interactions with uncertain and dynamic environments [1]. These interactions lead to changes in gene expression [2,3] and subsequently to the emergence of behaviours [4], in an attempt to exploit and/or fill an ecological niche [5]. An example of this process can be observed in eusocial insects. Development of these insects into castes possessing different morphology and physiology is determined by environmental exposure to cues such as nutrition, pheromones and temperature [6]. This allows a colony for example to balance caste sizes to fit the current ecological niche [5,6].

However, other organisms do not possess the same rigid form of epigenetic adaptation in regards to morphology and physiology. A broad range of research has shown that exposure to environmental stimuli during specific developmental periods can lead to changes in morphology and physiology, which can on occasion be passed down to other generations without changes to underlying DNA [7,8]. For example, a number of studies (for instance [9,10,11]) showed positive correlation between exposure to stress during prenatal or childhood periods and risk-averse behaviours in later life. Evidence has also been found, among other phenomena, of changes towards a more defensive phenotype following exposure to new predators [12], and of links between parental malnourishment during prenatal development and energy conservation (sedentary behaviour and weight retention) in later life [13].

While it is possible to measure and observe the effects of exposure to different environmental stimuli on development, there is still uncertainty over the actual mechanisms. Recent studies have however shown evidence of hormones

providing a signal for epigenetics [2,15] and that this mechanism can also be used in epigenetic robotics [14]. This is backed up by studies into prenatal development where exposure and absorption of hormones originating from the mother [8] and sibling (multi pregnancy) [15] or postnatal hormonal injects [16] lead to changes in gene expression. Under this premise, when hormones in the bloodstream reach tissues with cognate receptors they have the potential to elicit changes in that cell. These changes are dependent on the size of the exposure and cell type [1,6] changing in physiology and morphology as discussed.

In this study we propose and test an epigenetic mechanism based on hormone signalling as a method to allow for greater adaptation of a robot to dynamic environments providing different opportunities and challenges. The hormone component that underpins the model and provides the catalyst for the epigenetic process has been adapted from [17]. In this model, hormones secreted in relation to homeostatic deficits and environmental cues, modulate the robot's action selection mechanism (ASM) in regards to resource perception and procurement. The study presented here has been framed within a multiple-environment, dynamic three-resource action selection problem, in which each set of experiments was conducted in varying ecologies requiring the robot to develop different behaviours to suit the current niche. To assess the performance of the robot, experiments were conducted with two variants of the ASM. The remainder of this paper is organised as follows. Section II presents the robotic architecture and environment. Section III provides an overview and results of a selection of experiments. Section IV provides a discussion and conclusion.

II. ROBOTIC MODEL

A. Environment

The environments varied depending on the set of experiments undertaken. However, their underlying characteristics remained constant. Each set took place within a 2mx2m bordered environment inhabited by a single robot. Each environment contained a number of energy and repair resources represented by two different coloured balls. A temperature or climate was also set for each environment and sensed internally by the robot.

B. Morphology

Experiments were conducted with a K-team Koala robot equipped with 16 infrared range finders, a camera and a laptop running the Ubuntu Linux operating system using the Code:Blocks C++ environment and the openCV libraries for image processing (see Fig. 1).

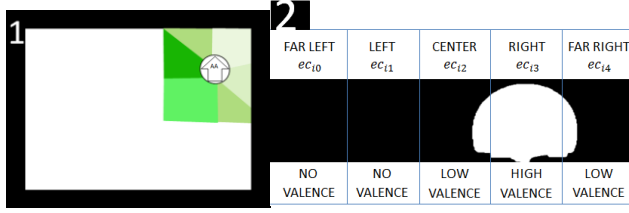


Fig. 1. Vision (Left, 1) and infrared (Right, 2) sensors. Left (1): Example of directional levels of perceived attractiveness of the environment (environmental cues) when close to a corner; the darker the colour the higher the attractiveness. Right (2): Detection of an environmental cue (ec_{ij}), where i is the type of cue and j is the position. Using the coordinates, cues are grouped into one of five directions—far left to far right. Both 1 & 2 underpin the motivations and behaviours discussed later.

Due to the lack of touch sensors, the infrared sensors play dual roles within the architecture. Firstly, placed into eight overlapping groups of four sensors (see fig 2), they are used to calculate the resistance and obstruction found in the eight cardinal or ordinal directions surrounding the robot. To encourage progressive movement, a slight bonus is given to the two most forward facing sensors.



Fig. 2 example of three of the eight overlapping sensor grouping, one group for each colour.

Secondly, in lieu of touch sensors, the infrared are used to extend the robot's body by creating a one-centimetre exclusion zone around its outline using a threshold. Any encroachment of this zone is registered as contact and measured in two parts: the force of initial impact and continued pressure.

C. Physiology

We have implemented a biologically-inspired robotic physiology consisting of three survival-related homeostatic variables (v_i), which must be maintained within a pre-set boundary ($0 \leq v_i \leq X$) for continued survival. The combinational deficits are further used to monitor the overall wellbeing in a manner similar to a viability zone [18]. The homeostatic variables in questions are energy, condition and temperature (see Table 1). While energy decreases at a constant rate equivalent to a basal metabolic rate, condition decreases in a semi-unpredictable manner with deficits occurring from collisions within the environment. These deficits can only be recovered by

finding specific resources within the environment. Temperature ($Rtemp$) is again semi-unpredictable and generated depending upon the current velocity (V) and climate. Cooling down (dissipation of temperature) occurs at a constant rate of cd ,

$$Rtemp += \frac{v_{\{current\}}}{v_{\{max\}}} climate - cd \quad (1)$$

Where $v_{\{current\}}$ is the current velocity and $v_{\{max\}}$ is the maximum velocity. Under normal climates, a moderate speed can be maintained without risk of overheating. However, a combination of constant higher speeds or hotter climates will quickly push temperature into the critical or lethal boundaries.

Table I. Homeostasis variables

H.Var	Ideal Value	Limit	Cause of deficit	Recover per action loop
Energy	100	0	$Bmr=0.1V$	1*
Cond	100	0	contact V	1*
Temp	0	100	$TempG\Delta$	$T=0.8$

* The robot must be near the resource for recover to commence

D. Hormone System

At the core of the ASM lies a hormone-like system inspired from models such as [17,20,24]. In the model used in this study, hormones (H) are separated into two distinct groups of endocrine hormones (Eh) and neurohormones (Nh) (see Table 2). The Eh group is made up of three hormones, one associated to each homeostatic variable. For each hormone, secretion occurs via a gland (g) and depends upon the current homeostatic deficit and the activity level of the gland (a_g),

$$s_g = x(a_g^{\text{deficit}}) \quad (2)$$

where x scales the size of secretion. Once released into a virtual bloodstream, each secretion persist within the system for a random number of actions loops (du) between a minimum and maximum ($min \leq du \leq max$) before decay occurs.

The second group, Nh consist of a solitary hormone D1, which facilitates aggressive behaviour. Unlike Eh, secretion is triggered from mean size of the external environmental cues (ec_{ij}) and disperses at a constant rate of δ ,

$$s = (s\delta) + x_i \bar{e} c a_g \quad (3)$$

where x is a predetermined weighting factor ($0 \leq x_i \leq 1$). Also different to the Eh model, a_i is not a set value. Instead, the activity level of the gland is stimulated by the mean concentration of the Eh hormones within the body in a similar manner to tropic hormones [23].

$$a_g = e^{-4e^{-0.5E\bar{H}}} \quad (4)$$

As the formula uses a Gompertz (sigmoid) function, with these variables $(-4, -0.5)$ the Nh 's a_g will lie between 0 and 1 depending upon the current hormone concentration (Eh). A value of 1 would indicate the gland being fully active while 0 signifies that the gland is inactive. The final part of the Nh equation models neuroreceptor sensitivity ($sens$),

$$Nh = e^{-3e^{sen \times s \times a_g}} \quad (5)$$

where sen the sensitivity of the receptor to the hormone, and $s \times a_i$ the level of hormone in the body.

Table II. Robotic hormones

H. Name	H. Type	Trigger	Ψ_i
E1	Eh	Engy deficit	0.09
C1	Eh	Cond deficit	
T1	Eh	Temp deficit	
D1	Nh	Visual cues	Varies

E. Hormones and the ASM

The ASM consists of a two-step computational layer voting-based architecture [27] where the action chosen is the one that provides the greatest valence for the overall system. The first layer of computation deals with the calculation of the current motivations (m_{ij}) (see Table 3). Unlike previous work that uses physiological deficits to provide the basis for drives (d_i) of motivations [17,20], in this model we use the hormone levels derived from the physiological deficits for this calculation.

Although three drives are present, we only need to directly calculate the intensity of hunger and damage. The hyperthermia drive which is satisfied from reduced or no movement instead suppresses other drives. In addition to the internal state, motivations are influenced by environmental cues by implementing part of an algorithm proposed in [17],

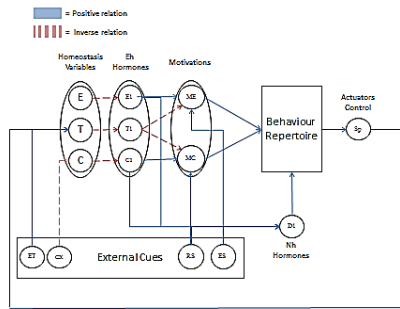


Fig. 3. Hormone-based Architecture.

$$m_{ij} = \begin{cases} \frac{d_i ec_{ij}}{1 + T1} & \text{if } d_i > mtT1 \\ 0 & \text{Otherwise} \end{cases} \quad (6)$$

where $mtT1$ is the activation threshold with mt being a constant value (0.75) ec_{i2} is given a +1 score to simulate a restless mechanism and allow forward movement without

external stimuli with E1 or C1 levels. To further reduce excessive switching of motivations, ec_{i2} is given a 10% bonus to its value as a form of "hysteresis".

Table III. Motivation types

Motivation	Drive	ext Stimuli	Suppressed by
Hunger (ME)	E1	Eng source	T1
Damage (MC)	C1	Rep source	T1
Hyperthermia	T1	Climate	E1 & C1

The second layer of the architecture calculates the behaviour to execute given the current motivational state and environmental conditions. Unlike previous hormone-based architecture such as [17, 20], no explicit behaviours have been modelled. In our case, as in [24], behaviours occur from dynamic combinations of different systems with no pre-set physiological cost or gain. The cost or gain of behaviour execution results from the sum of physiological change that occurred during the action.

The different systems that make up behaviours can be thought of as characteristics of the robot. The first of these systems is the need for personal space (Ps) [21], an area that the robot will treat almost as an extension of its own body. Using a similar technique as with the "extended body" (the IR-based touch sensors around the robot's body), the robot will normally maintain the Ps free from other objects. The radius of the Ps zone is determined by the current C1 hormone concentration. Encroachment will lead to attempts to re-establish a space by moving along the path of least resistant (d_n), with a slight preference to going forward. D1 counteracts the tendency to maintain the Ps free, allowing the tolerance of encroachment while trying to satiate drives. At high levels, D1 will facilitate physical contact,

$$Ps = \left(ns + ns \frac{C1}{Eh_{\{max\}}} \right) (1 - D1) \quad (7)$$

where ns is the normal, unadjusted size of personal space desired and $Eh_{\{max\}}$ the highest potential concentration of the Eh hormones. Fig. 2 provides an overview of the architecture used.

F) Hormone-Signalled Epigenetics

The final and following system introduces an epigenetic adaptation mechanism into the architecture. Taking inspiration from recent biological studies [2,4], hormones provide the driving force behind epigenetic change. In our robot, hormone level indirectly and directly provides a fairly accurate measure of current conditions in the environment and level of situatedness. For instance, a high E1 is an indication of issues maintaining homeostatic balance in terms of energy. Combined with the concentration of D1, it is possible to determine the root of the imbalance, either the scarcity of, or difficulty of access to the resources.

Let us look at an example from a preliminary experiment, in which the environment contained an abundance of fast moving energy sources. The robot's speed, which is constrained by temperature, was naturally around half of the sources. Unless a source could be trapped in a corner, the robot was limited to "taking bites" when passing nearby, soon leading to homeostatic imbalance. Eventually, the deficits and external stimuli increased the concentration of N1 and E1 to the point where the robot would tolerate increased temperatures caused by the needed increase of speed.

Although the robot was able to successfully restore homeostatic balance in the above example, inefficiency was prevalent. This was primarily due to the growth of energy deficits to dangerous levels before hormone concentration could stimulate the required speed to facilitated feeding.

An argument could be made that inefficiencies in behaviour were caused due to poor calibration of the variables that regulate the hormone system. While potentially possible to predetermine passable weightings for a four-hormone system in a simplistic environment, even in simple systems this has been shown to lead to unintended behaviour [26]. Trying to do so for a more complex architecture in fully dynamic real world environments would almost certainly end up as a deterministic chaotic behaviour [25]. We are thus unable to correctly predict the needed initial state to achieve the desired behaviour in dynamic environments.

An alternative, which we demonstrate here, is to implement epigenetic adaptation, whereby development of the glands that secrete hormones (formula 8) and sensitivity of receptors that receive them (formula 9) are influenced by the external environment. In this implementation, the epigenetic process will only occur during the early life of the robot, currently defined as the first three minutes

$$\phi_i += \frac{Eh_i}{k} \quad (8)$$

$$sen- = \frac{(D1-0.5)}{k} \quad (9)$$

where k is a predetermined constant value which regulates the speed of epigenetic change (the value of k is different for 13,14).

The epigenetic robot was put through exactly the same preliminary experiment. Unlike the original robot, which "chased" energy sources until eventually gaining enough speed to catch them, the epigenetic model developed tailored behaviours that exploited the current environment. Specifically, the robot developed behaviours where it would remain sedentary until an energy source passed closely, at which point the robot would "pounce", giving chase at full speed and often pinning the source to a wall until finished feeding (see Fig. 4). Such behaviour presents structural and functional similarities with ambush predator-like behaviours in animals.

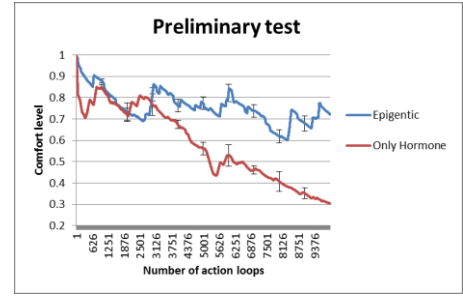


Fig. 4. Preliminary experiment results. Comfort level provides a measure of wellbeing described in greater detail in the next section (Section III).

III. EXPERIMENTS

A total of 35 runs split amongst three scenarios of 10, 10, 15 respectively were tested. Each run lasted 10,000 steps with 16 steps per second, equivalent to around 10 minutes 40 seconds per run. The epigenetic system was active during the first 3 minutes (2880 steps). A second set of runs was conducted in the same manner for a robot without the epigenetic mechanism to serve as a base for comparison. We used the same viability-based performance metrics as in [17], particularly the viability indicators called "comfort", which is derived from the average homeostatic deficit at any time, and "risk of death", which indicates how close the internal state is from reaching lethal values.

A. Scenario one

The first scenario took place within a walled environment, devoid of any features save one of each of the resources placed in opposite corners with a low temperature. As can be seen in Fig. 5 both robots performed in an adequate manner. Since the environment was devoid of any real challenge apart from finding and consuming the correct resource and the right time, the only real danger came from running out of energy. The performance of both robots demonstrates that internal variables are all within an adequate range. One of the more interesting behaviours to emerge was tight circling around resources. This behaviour occurred when E1/C1 lagged behind actual deficits, leading to a tendency to find the resource but not to consuming it. This behaviour was observed more often in the epigenetic model due to its adaptive nature and is likely to be the cause of its better performance during the later periods.

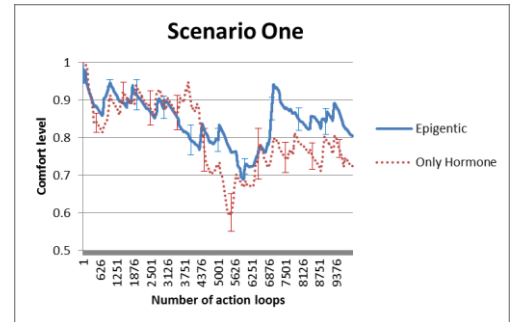


Fig. 5. Results from scenario one.

B. Scenario Two

The second scenario featured the same environment as the first, save the presence of light movable objects. These were placed around the resources, leaving gaps for the robot to see what was contained behind. This scenario presented two challenges. Firstly, partially hidden resources led to smaller environmental cues. Secondly, in order to reach the resource, the robot needed to push aside the object, causing both an encroachment of Ps and loss of condition caused from contact. This led to a further issue, namely a need for increased emphasis on maintaining condition so that obstacles could be safely moved without the risk of death. Once the robot had left the immediate vicinity, the objects were replaced to their original standing point. Fig.6 shows the results for this scenario.

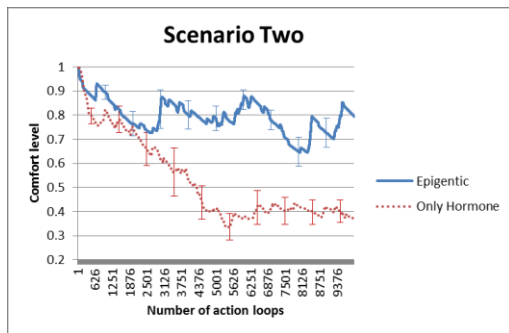


Fig.6. Results from scenario two.

Under this scenario, the non-epigenetic robot performed particularly poorly. Although in one run the robot ended with an average comfort level of 0.72, in three other cases the robots died around the half-way stage, which would explain the high standard deviation. Predictably, the non-epigenetic robot was unable to adapt to a harsher environment. On the one hand, failing to maintain enough condition to force through to the resource, it would often patrol the perimeter looking for a non-existent entrance. Added to this, a smaller Ec led to lower levels of D1, further reducing the ability to access resources. The deaths came from two cases of starvation and one of a fateful “last-ditch effort” run.

In comparison, the epigenetic model was able to adapt to the new environment relatively successfully. Perception of resources that were visible but inaccessible led to a chain reaction of internal changes. Since the robot possessed neuroreceptors with increased sensitivity due to high D1 hormone levels, “aggressive” behaviour emerged with less stimulation. This in turn permitted the robot to push past obstacles earlier. Doing so and creating condition deficits simulated the growth of the C1 gland, allowing the hormone to be produced at greater rates, which in turn increased the tendency to replenish condition earlier.

C. Scenario Three

The final scenario tested was slightly different from the first two. In this scenario, we examined how the architectures would be able to cope with uneven resource density within the environment. In order to do this, three sets of five runs were conducted. The first set featured four additional energy sources, giving a total of five. The second set featured four additional repair sources. The final set featured a hot temperature. Excluding the aforementioned changes, each set was the same as the first scenario. An overview of the combined results is shown in Fig.7

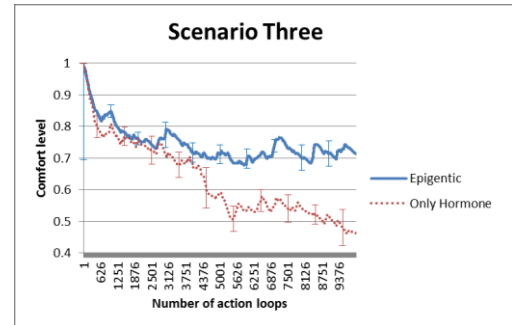


Fig.7. Combined Results from scenario three.

The first two sets provided similar problems for the robot’s architecture. Firstly, the additional stimuli of excess resource could potentially pose a distraction while searching for the rarer resources. Secondly, resources are semi-dynamic and can be moved when touched, potentially blocking the view of the rare resource. In order to be successful within this environment, the robot would need to be more sensitive to resources and act in a timely manner when the opportunity to replenish was present.

Even with the hormone system, the non-epigenetic model suffered in dealing with uneven resources. The overload of stimulation from the excess resources meant that the homeostatic deficit correlating to the rare resources was on average 32 per cent greater. The two occasions when the rare resource was hidden behind another resource was also extremely problematic as the robot would consistently stop searching for it in order to satisfy the other motivation.

In contrast, the epigenetic model was able to successfully adapt in a similar manner to the prior scenario. Early homeostatic deficits led to the growth of an associated hormonal gland. This in turn led the robot to being more sensitive to these deficits, filtering out the distraction and noise of the other resources. Under the epigenetic model, when both stimuli were present the deficit of the overabundant resource would need to be around 42 per cent greater in order to attract the robot’s attention. Further, in the four cases when the rare resource was hidden the deficit for the common resource would need to drop to 60-80 per cent before the search for the rare resource was broken.

Finally, the set with the high level of temperature did not yield considerable difference in terms of comfort level.

While the epigenetic model did perform better in the experiments, the biggest difference was qualitative in terms of behaviours that emerged from the models. The epigenetic model developed a behaviour which could be considered as stalking-like. The robot would move toward the resource taking breaks until it was within a certain range, at which point it would burst forward at full speed. In contrast, the hormone model would maintain a slower constant speed and move towards the desired resource. While both behaviours worked, the epigenetic approach generated less overall heat.

IV. CONCLUSION

Environmental stimuli have been shown to lead to physiological and morphological changes in biological organisms. These changes modify the underlying behaviour of the organism in order to adapt to its current environment. Epigenetic adaptation can be broadly categorised in one of two groups. First, epigenetic changes occur as an on-off switch for gene expression such as in [5,6,7], where there is mostly a predictable standardised outcome. Second, epigenetic changes can also occur through subtle changes in neural and endocrine systems such as [9,10,11,12]. Under this latter approach, the behaviour of organisms with the same underlying DNA can vary significantly when exposed to the same stimuli [22].

In this paper we have looked at introducing the latter form of epigenetic adaptation in an autonomous robot. This adaptation process, signalled by hormones, allows the robot to organise its internal systems to be in tune with its external environment. As we have shown in the tested scenarios, this process significantly improves the robot's adaptability when placed into environments containing different dynamics. It could perhaps be argued that the tests show some bias toward the epigenetic model, as the hormone-only model has not been tailored to the environment as is the usual case in robotic experiments [20]. We counter this argument by stating that this is precisely the purpose of our system. What we are investigating is how to develop robot architectures that have the potential to enter into an unknown environment and not only survive but develop behaviours that allow the robot to exploit its current niche and "thrive".

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References

- [1]. R Jaenisch, A Bird. Epigenetic regulation of gene expression: how the genome integrates intrinsic and environmental signals. *Nat Genet* 2003; 33(Suppl): p.p 245–254
- [2]. AL fowden AJ Forhead. Hormones as epigenetic signal in developmental programming. *Exp Physiol*. 2009 Jun; p.p 607-25.
- [3]. X Zhang and S M Ho. Epigenetics meets endocrinology *Journal of Molecular Endocrinology* 2011 46, p.p 11–32
- [4]. D Crews. Epigenetics and its implications for behavioural neuroendocrinology. *Frontiers in Neuroendocrinology* 29 2008 p.p 344-357
- [5]. C Narain. Changing behaviour with epigenetics, *Natural Neuroscience* 15, 2012 p.p 1329
- [6]. A Chittka, Y Wurm, L Chittka Epigenetics: The Making of Ant castes, *Current Biology* Vol 22 No 19 2012 p.p 836-838
- [7]. F Lyho, C Beisel, J Marhold, R Paro, Epigenetic regulation in *Drosophila*. *Curr Top Microbiol Immunol* 310. 2006 p.p 23-44. .
- [8]. D Crews et al. Transgenerational epigenetic imprints on mate preference. *Proceedings of the National Academy of Sciences* 104. 2007 p.p 5942–5946
- [9]. S. Maccaria, et al. Prenatal stress and long-term consequences: implications of glucocorticoid hormones *Neurosci Biobehav Rev*. Jan-Mar;27(1-2). 2003 p.p 119-27
- [10]. T F Oberlander, et al. Prenatal exposure to maternal depression, neonatal methylation of human glucocorticoid receptor gene (*NR3C1*) and infant cortisol stress responses. *Epigenetics*. 2008;3:97–106.
- [11]. H Karst, S Berger, G Erdmann, G Schutz, M Joëls. Metaplasticity of amygdalar responses to the stress hormone corticosterone. *Proceedings of the National Academy of Sciences*.2010;107: p.p 14449–14454
- [12]. H Miyakawa et al, Gene up-regulation in response to predator kairomones in the water flea, *Daphnia pulex*. *BMC Dev Biol* 2010 p.p 10- 45.
- [13]. M H Vickers, B H Breier, W S Cutfield, P L Hofman, P D Gluckman. Fetal origins of hyperphagia, Obesity, and hypertension and postnatal amplification by hypercaloric nutrition. *Am J Physiol Endocrinol Metab* 2000, p.p 83-87
- [14]. D Bowes, R. Adams, L. Cañamero, V. Steuber, and N. Davey. Should I worry about my stressed pregnant robot? In *Proc. EpiRob*, 2009.
- [15]. L Clemens, B Gladue, L Coniglio. Prenatal endogenous androgenic influences on masculine sexual behaviour and genital morphology in male and female rats. *Hormones and Behavior* volume 10 issue 1. 1978 p.p 40-53
- [16]. K T Brill et al. Single and combined effects of growth hormone and testosterone administration on measures of body composition, physical performance, mood, sexual function, bone turnover, and muscle gene expression in healthy older men. *J Clin Endocrinol Metab* 87: 2002 p.p 5649–5657
- [17]. O Avila-García, L Cañamero. Using hormonal feedback to modulate action selection in a competitive scenario. In: *From Animals to Animats: Proceedings of the 8th International Conference of Adaptive Behavior (SAB'04)*, Cambridge, MA, MIT Press 2004 p.p 243–252.
- [18]. W R Ashby. *Design for a Brain*, Chapman and Hall, London (1952).
- [19]. D, McFarland. Experimental Investigation of Motivational State. In *Motivational Control Systems Analysis*, D. McFarland, ed., 251–282. London: Academic Press 1974
- [20]. J.L Krichmar "A biologically inspired action selection algorithm based on principles of neuromodulation" in *The 2012 International Joint Conference on neural networks* 2012 p.p 1-8
- [21]. E T Hall. *The Hidden Dimension*. Anchor Books 1966
- [22]. A S Clark, D J Wittwer, DH ABBOT, ML Schneider. Long-term effects of prenatal stress on HPA axis activity in juvenile rhesus monkeys *Dev Psychobiol*. 1994 p.p 257-269
- [23]. L Sherwood *Human Physiology: From Cells to Systems* Wadsworth Publishing Co Inc; 5th Revised edition edition 2003
- [24]. R L B French, L Cañamero. Introducing Neuromodulation to a Braitenberg Vehicle. In: *Proc. 2005 IEEE Int Conf on Robotics and Automation: ICRA*. 2005 p.p. 4188-4193.
- [25]. M Marek, Chaotic behaviour of deterministic dissipative systems. Cambridge University Press 1991
- [26]. M Maris, R Boekhorst. Exploiting physical constraints: heap formation through behavioral error in a group of robots. In *Proc. IEEE/RSJ IROS'96*, pp. 1655-1660
- [27]. O Avila-García, L Cañamero, R Boekhorst. Analyzing the performance of 'Winner-Take-All' and 'Voting-Based' action selection policies within the two-resource-problem. In *Proc. 7th European conference in artificial life (ECAL03)*, 2003 pp. 733–742